

HW3 — Modeling (CRISP-DM 4)

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Project: Real-Time Human Activity Recognition from Smartphone Sensors.

Models trained:

two classical baselines on a reduced 6-feature set (Logistic Regression, Random Forest)

four deep models on the raw 128×9 signal (Main, Patch and CNN-Transformer, and the dual-branch **Fourier+Transformer**), plus an architecture ablation and a sensor ablation.

4.1 Select Modeling Technique

Output — Modeling Technique

We treat HAR as supervised multi-class classification of 2.56 s windows into six activities. We use two complementary techniques on two representations of the same windows:

- **Classical models on 6 engineered features.**
- **Logistic Regression** — linear, transparent reference.
- **Random Forest** — non-linear tree ensemble.
- **Transformers on the raw signal** (learn their own features via self-attention):
- **Main Transformer** — per-timestep tokens, sinusoidal positional encoding, CLS pooling.
- **Patch Transformer** — 8 timesteps per token (patch_size = 8).
- **CNN-Transformer** — 1-D conv front-end feeding the Transformer.
- **Fourier+Transformer** — dual-branch: one encoder over the time signal, one over the

per-channel FFT log-magnitude spectrum, fused before classification (our flagship model).

This lets us compare feature-based classical ML against deep models on the raw waveform, and test whether explicit frequency information helps.

Output — Modeling Assumptions

- Each 2.56 s window (128 samples × 9 channels) is an independent sample.
- Train/test subjects never overlap → scores estimate generalisation to new people.
- The 6 features are standardised (train statistics); the raw signal is z-scored per channel (train statistics) so all channels share a scale.

- 2.56 s holds enough motion/periodicity to identify the activity; labels are reliable.

4.2 Generate Test Design

Output — Test Design

- **Split** : Official UCI subject-disjoint partition: 7,352 training windows (21 subjects) and **2,947 test windows (9 unseen subjects)**, used once for final evaluation.

- **Feature reduction (classical track)**. We keep the 6 features with the highest mutual

information with the label, computed on the **training set only**:

tBodyAcc-max()-X

tGravityAcc-min()-Y

tBodyAccJerk-max()-X

tGravityAcc-max()-Y

tBodyAccMag-max()

tGravityAccMag-max().

These combine movement-intensity features

- **Validation (deep track)**. From the training pool we hold out 5 subjects (1,764 windows) as a

subject-wise validation set. models train on the remaining 5,588 windows, and we keep the best

validation-macro-F1 checkpoint. The validation set drives all architecture choices (the ablation).

- **Metrics**

macro-F1 , accuracy, per-class precision/recall/F1, confusion matrices, and ECE (calibration).

Fixed seed = 42.

4.3 Build Model

Output — Parameter Settings

- **Classical:**

LogReg C = 2.0, max_iter = 2000 (+ StandardScaler)

Random Forest 600 trees , min_samples_leaf = 2

- **Transformers (shared core):**

Dimension model = 64, 4 heads, 2 layers, feed-forward 128, dropout 0.15 Sinusoidal PE, CLS pooling. Optimiser AdamW (lr 1e-3, weight decay 1e-4), cosine schedule, batch 128, gradient clipping 1.0, 20 epochs, best-validation checkpoint.

Fourier+ Transformer =c136,326 parameters.

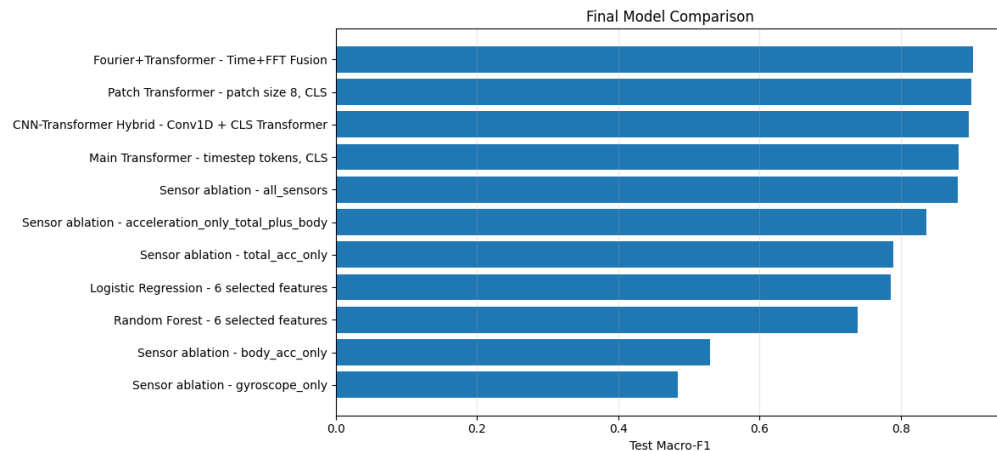
Output — Models

Eight trained models: 2 classical (6 features), 4 deep (raw signal), plus 7 ablation variants and 5 sensor-group variants (validation/secondary studies).

Output — Model Description

Final **test-set** performance (subject-disjoint, 2,947 windows):

Model	Input	Test acc	Macro-F1
Fourier+Transformer (time + FFT)	raw 128×9	0.905	0.902
Patch Transformer (patch 8)	raw 128×9	0.901	0.899
CNN-Transformer hybrid	raw 128×9	0.896	0.896
Main Transformer (per-timestep)	raw 128×9	0.883	0.882
Logistic Regression	6 features	0.787	0.785
Random Forest	6 features	0.742	0.739



Final model comparison by test macro-F1.

- **Fourier+Transformer is best** (macro-F1 0.902) and notably fixes the hard static postures: its

SITTING F1 = 0.899 and STANDING F1 = 0.909 vs. 0.803 / 0.839 for the Main Transformer — the FFT branch separates near-static activities.

- **Patch tokenisation helps** ($0.899 > 0.882$): fewer, less-noisy tokens generalise better.
- The **CNN front-end** is competitive (0.896). The **Main Transformer** is the weakest deep model and is well-calibrated ($\text{ECE} = 0.048$).
- The **6-feature classical models** (0.785 / 0.739) are a deliberately minimal baseline: six numbers capture coarse structure but not fine within-group distinctions.
- Across every model, **LAYING is perfect** and no dynamic activity is confused with a static one, all error is within the walking group or within the sitting/standing pair.

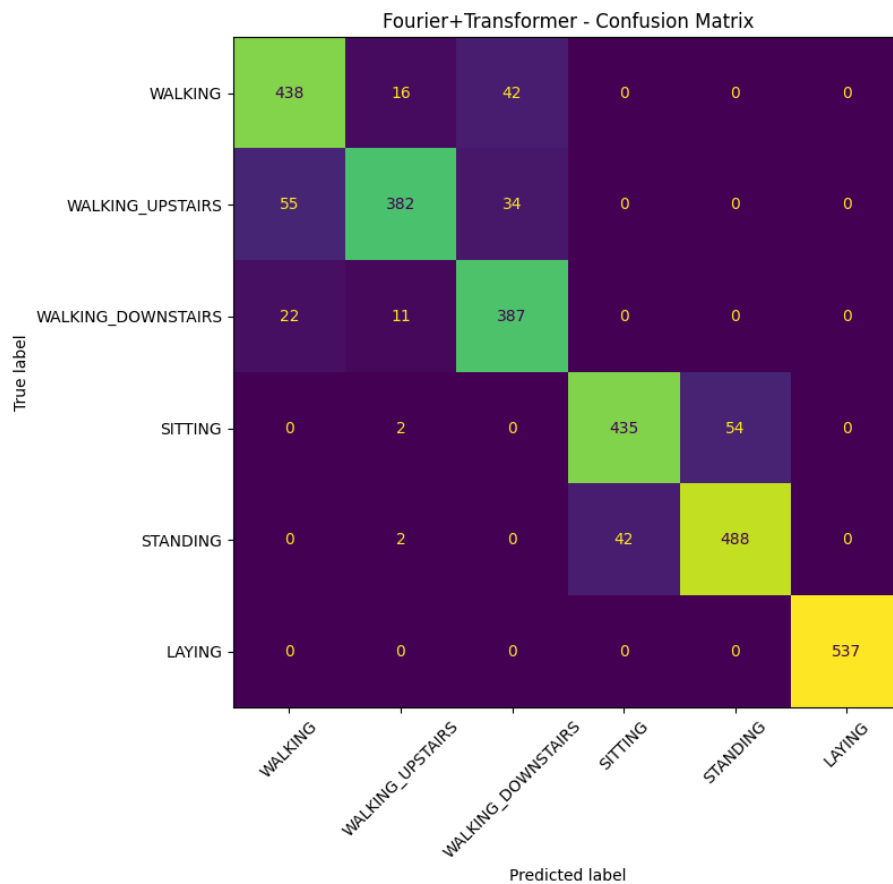
Difficulty: the Transformers overfit the training subjects (train F1 ~ 0.96 – 0.98 vs. test ~ 0.88 – 0.90), reflecting between-subject variability.

4.4 Assess Model

Output — Model Assessment

The ranking above (best $\rightarrow$ worst by macro-F1) puts the four Transformers ahead of the 6-feature baselines, with the Fourier+Transformer first.

Confusions : The only meaningful errors are **SITTING $\leftrightarrow$ STANDING** and mix-ups among the three walking variants. LAYING is always correct. The Fourier+Transformer's frequency branch is what shrinks the SITTING/STANDING confusion.

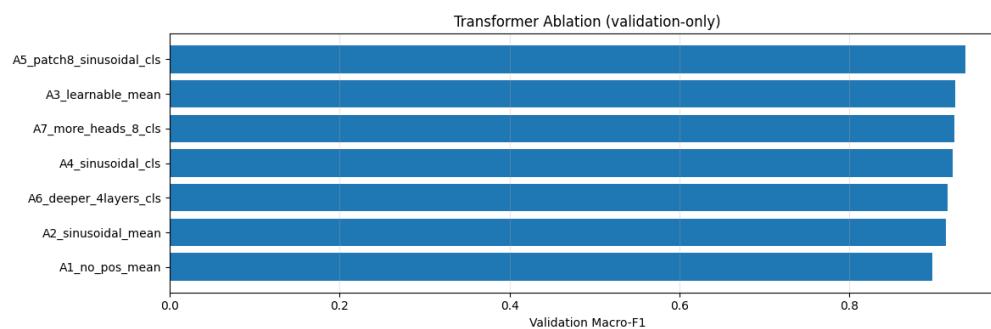


Fourier+Transformer — test confusion matrix.

Per-subject. Accuracy varies strongly by person (0.64 for subjects 9–10 up to 1.00 for subject 24) a few hard subjects drive most of the error the main generalisation limitation.

Ablation (validation macro-F1): best is **patch 8 + sinusoidal PE + CLS (0.937)**.

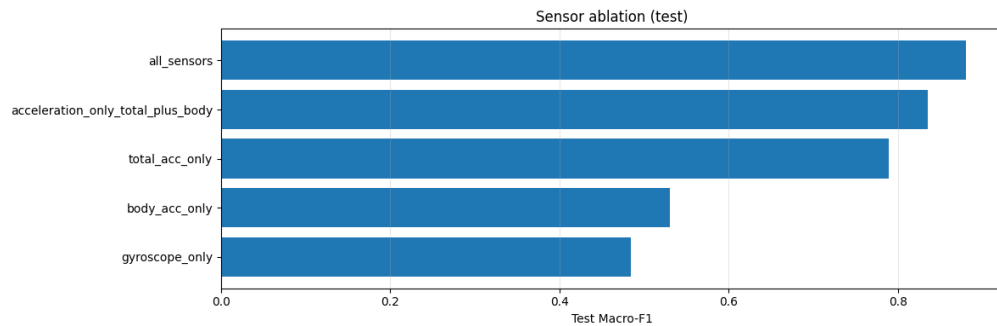
Removing positional encoding is worst (0.897). extra depth/heads add little. CLS $\geq$ mean pooling.



Transformer architecture ablation (validation macro-F1).

Sensor ablation (test macro-F1): all 9 sensors 0.881. total-acc only 0.789. gyroscope only

0.484. The **accelerometer carries most of the signal**, the gyroscope alone is weak, and using all sensors is best , answering question about which signals matter.



Sensor-group ablation (test macro-F1).

Against the success criteria: a working pipeline, full multi-class metrics, and clear separation of the core activities met by the deep models (~0.90).

Output : Revised Parameter Settings

- Adopt **patch tokenisation + CLS + sinusoidal PE** as the default recipe, and the

Fourier+Transformer as the final model (most accurate, most robust on the hard classes).

- Residual error sits in SITTING/STANDING and a few subjects, not in any single hyper-parameter, so

further grid search has limited upside.

- Next iteration: light regularisation / early stopping to close the overfitting gap, data augmentation (jitter, rotation) for cross-subject robustness, and if a stronger classical baseline is wanted — relaxing the 6-feature limit. This closes the build → assess loop.